

■ Expense Tracker MCP Server

Complete Learning & Implementation Guide | FastMCP, Neon PostgreSQL, Cloud Deployment, and uv vs pip

1. Project Overview & Architecture

This project implements a multi-user **Model Context Protocol (MCP)** server that connects AI clients (like Claude Desktop and Claude.ai) to a serverless **Neon PostgreSQL** database. It provides full expense management capabilities with strict user data privacy.

AI Client (Claude / Agent)	→	FastMCP Server (server.py)	→	Neon PostgreSQL (db.py)
Prompts & Tool Calls (Bearer Auth Header)		Resolves user identity Validates arguments		Stores api_keys & expenses Enforces tenant scoping

2. Neon PostgreSQL Database Schema

The database uses PostgreSQL with pgcrypto for UUID generation. Execute this SQL in your Neon Console SQL Editor:

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;

CREATE TABLE IF NOT EXISTS api_keys (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     uuid NOT NULL DEFAULT gen_random_uuid(),
  api_key     text NOT NULL UNIQUE,
  label       text,
  created_at  timestampz NOT NULL DEFAULT now(),
  revoked     boolean NOT NULL DEFAULT false
);

CREATE TABLE IF NOT EXISTS expenses (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     uuid NOT NULL,
  amount      numeric(12,2) NOT NULL CHECK (amount > 0),
  category    text NOT NULL,
  description  text DEFAULT '',
  expense_date date NOT NULL DEFAULT current_date,
  created_at  timestampz NOT NULL DEFAULT now(),
  updated_at  timestampz NOT NULL DEFAULT now()
);

CREATE INDEX IF NOT EXISTS idx_expenses_user_date ON expenses (user_id, expense_date DESC);
CREATE INDEX IF NOT EXISTS idx_expenses_user_category ON expenses (user_id, category);
CREATE INDEX IF NOT EXISTS idx_api_keys_key ON api_keys (api_key);
```

3. Step-by-Step Local Setup Guide

• **Step 1: Clone & Navigate:**

```
git clone https://github.com/sonawanewdinesh18/expense-tracker-mcp-server.git
```

• **Step 2: Create Environment & Install:**

```
python -m venv .env | .env\Scripts\Activate.ps1 | pip install -r requirements.txt
```

• **Step 3: Set Environment Variables in .env:**

```
DATABASE_URL=postgresql://user:pass@ep-xyz.aws.neon.tech/neondb?sslmode=require
```

(Ensure no surrounding quotation marks!)

- **Step 4: Create Your User API Key:**
python create_user.py "Your Name" → Prints your unique user_id and api_key (e.g. et_...).

4. Deep Dive: 'uv' vs Traditional 'pip'

What is uv? uv is an ultra-fast Python package and project manager developed by Astral (written in Rust). FastMCP tutorials frequently reference uv because it eliminates manual virtual environment activation and installs packages 10-100x faster.

Workflow Task	Standard pip / venv	Modern uv Way ■
Create Virtual Env	python -m venv .venv .venv\Scripts\Activate.ps1	Automatic (handled by uv)
Install Dependencies	pip install -r requirements.txt	uv add fastmcp "psycpg[binary]"
Run Server	python server.py	uv run server.py
Run FastMCP Inspector	mcp dev server.py	uv run mcp dev server.py
Create User	python create_user.py "Alice"	uv run create_user.py "Alice"

5. Cloud Deployment (FastMCP Cloud / Prefect Horizon)

- **1. Push Repository:** Push your code to GitHub (branch main).
- **2. Connect Horizon:** Log in to **horizon.prefect.io** and select your repository.
- **3. Entrypoint:** Set Server Entrypoint to server.py:mcp.
- **4. Environment Variables:** Set DATABASE_URL to your clean Neon PostgreSQL connection string.
- **5. Deploy:** Your server goes live at https://your-server.fastmcp.app/mcp.

6. Connecting to Claude (Claude.ai & Claude Desktop)

Claude.ai Custom Connectors: Add custom connector ➡■ Enter URL https://your-server.fastmcp.app/mcp ➡■ Select *Always required & No client ID (automatic)* ➡■ Authorize popup.

Claude Desktop: Add to claude_desktop_config.json under mcpServers with your server URL and Authorization header.

7. Ready-to-Use Test Prompts for Claude

Goal	Prompt to send to Claude
Add Expense	I spent \$45.50 on groceries and \$12.00 for lunch today. Record both under Food.
List & Filter	Show me a list of all expenses recorded under the Food category.
Summary	How much have I spent so far this month? Give me a breakdown of totals by category.
Update	Change my lunch expense amount from \$12.00 to \$15.50.
Monthly Report	Run my monthly expense report and give me 3 practical tips to optimize my spending next month.